

# LABORATORY RECORD

**Course:** Full Stack Web Development

**Course Code:** 23CM4121

**Experiment No.:** 8

**Student Name:** N. Samhavya

**Roll No.:** A24126552099

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## Title

**Implementation of HTTP Methods using JSON Server**

## Aim

To implement and demonstrate the **GET, POST, PUT and DELETE HTTP methods** using JSON Server for managing data in a RESTful web application.

## Objectives

1. To understand the concept of a REST API and JSON Server.
2. To create a simple JSON-based REST API.
3. To perform **GET, POST, PUT and DELETE** operations.
4. To understand how client applications communicate with a server using HTTP methods.
5. To test API endpoints using a suitable tool such as **Postman** or a web browser.

## Theory

**JSON Server** is a lightweight tool that allows developers to create a REST API from a JSON file without developing a complete backend server.

It uses HTTP methods to perform operations on resources:

| HTTP Method | Purpose       | Example          |
|-------------|---------------|------------------|
| GET         | Retrieve data | Get all students |

|               |                      |                        |
|---------------|----------------------|------------------------|
| <b>POST</b>   | Add new data         | Add a student          |
| <b>PUT</b>    | Update existing data | Update student details |
| <b>DELETE</b> | Remove data          | Delete a student       |

For this experiment, a . .   file is used as the database.

Example resource:

```
{
  "students": [
    {
      "id": 1,
      "name": "Ravi",
      "branch": "CSE",
      "marks": 85
    }
  ]
}
```

When JSON Server is started, it automatically provides REST endpoints such as:

```
GET      /students
GET      /students/1
POST     /students
PUT      /students/1
DELETE   /students/1
```

Thus, JSON Server helps students understand RESTful API operations before working with a full backend framework.

     :

```
<!DOCTYPE html>
<html lang="en">
```

```
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-
scale=1.0">

  <title>Student Management System</title>

<style>
  body {
    font-family: Arial, sans-serif;
    margin: 40px;
    background-color: #f4f4f4;
  }

  h1 {
    text-align: center;
  }

  .form-container {
    background: white;
    padding: 20px;
    margin-bottom: 25px;
    border-radius: 8px;
  }

  input {
    padding: 10px;
    margin: 5px;
  }

  button {
    padding: 9px 14px;
    margin: 5px;
    cursor: pointer;
  }
```

```
table {
    width: 100%;
    background: white;
    border-collapse: collapse;
}

th,
td {
    padding: 12px;
    border: 1px solid #ccc;
    text-align: center;
}

th {
    background-color: #333;
    color: white;
}
</style>
</head>

<body>

<h1>Student Management System</h1>

<div class="form-container">

    <h2>Add Student</h2>

    <input type="text" id="name" placeholder="Student Name">

    <input type="text" id="branch" placeholder="Branch">

    <input type="number" id="marks" placeholder="Marks">

    <button onclick="addStudent()">Add Student</button>

    <button onclick="getStudents()">Refresh</button>

</div>
```

```
<div class="form-container">

  <h2>Student Records</h2>

  <table>

    <thead>
      <tr>
        <th>ID</th>
        <th>Name</th>
        <th>Branch</th>
        <th>Marks</th>
        <th>Actions</th>
      </tr>
    </thead>

    <tbody id="studentTable">
    </tbody>

  </table>

</div>

<script src="script.js"></script>

</body>

</html>
```

## script.js

```
const API_URL = "http://localhost:3000/students";

// GET: Display all students
async function getStudents() {

  try {

    const response = await fetch(API_URL);
```

```

const students = await response.json();

const table = document.getElementById("studentTable");

table.innerHTML = "";

students.forEach(student => {

    const row = document.createElement("tr");

    row.innerHTML = `
        <td>${student.id}</td>
        <td>${student.name}</td>
        <td>${student.branch}</td>
        <td>${student.marks}</td>

        <td>
            <button onclick="updateStudent('${student.id}')">
                Update
            </button>

            <button onclick="deleteStudent('${student.id}')">
                Delete
            </button>
        </td>
    `;

    table.appendChild(row);
});

} catch (error) {

    console.error("Error:", error);

    alert("Unable to fetch student records.");

}
}

```

```
// POST: Add a new student
async function addStudent() {

    const name = document.getElementById("name").value;
    const branch = document.getElementById("branch").value;
    const marks = document.getElementById("marks").value;

    if (name === "" || branch === "" || marks === "") {

        alert("Please enter all student details.");

        return;
    }

    const student = {

        name: name,
        branch: branch,
        marks: Number(marks)

    };

    try {

        const response = await fetch(API_URL, {

            method: "POST",

            headers: {

                "Content-Type": "application/json"

            },

            body: JSON.stringify(student)

        });

        if (response.ok) {

            alert("Student added successfully.");
        }
    }
}
```

```

        document.getElementById("name").value = "";
        document.getElementById("branch").value = "";
        document.getElementById("marks").value = "";

        getStudents();

    }

} catch (error) {

    console.error("Error:", error);

    alert("Unable to add student.");

}

}

// PUT: Update student
async function updateStudent(id) {

    try {

        // Get current student details
        const response = await fetch(`${API_URL}/${id}`);

        const student = await response.json();

        // Ask for updated values
        const name = prompt("Enter student name:", student.name);

        if (name === null) {
            return;
        }

        const branch = prompt("Enter branch:", student.branch);

        if (branch === null) {
            return;
        }
    }
}

```

```

const marks = prompt("Enter marks:", student.marks);

if (marks === null) {
    return;
}

const updatedStudent = {

    name: name,
    branch: branch,
    marks: Number(marks)

};

// Update using PUT
const updateResponse = await fetch(`${API_URL}/${id}`, {

    method: "PUT",

    headers: {
        "Content-Type": "application/json"
    },

    body: JSON.stringify(updatedStudent)

});

if (updateResponse.ok) {

    alert("Student updated successfully.");

    getStudents();

}

} catch (error) {

    console.error("Error:", error);

```

```

        alert("Unable to update student.");
    }
}

// DELETE: Delete student
async function deleteStudent(id) {

    const confirmDelete = confirm(
        "Are you sure you want to delete this student?"
    );

    if (!confirmDelete) {
        return;
    }

    try {

        const response = await fetch(`${API_URL}/${id}`, {

            method: "DELETE"

        });

        if (response.ok) {

            alert("Student deleted successfully.");

            getStudents();

        }

    } catch (error) {

        console.error("Error:", error);

        alert("Unable to delete student.");

    }
}

```

```
}
```

```
// Load students when page opens  
getStudents();
```

## OUTPUT:

The screenshots show a web application titled "Student Management System" running on `http://localhost:5500/index.html`. Each screenshot includes an "Add Student" form with fields for "Student Name", "Branch", and "Marks", and buttons for "Add Student" and "Refresh". Below the form is a "Student Records" table with columns: ID, Name, Branch, Marks, and Actions (Update, Delete).

**Screenshot 1 (Top Left):** Initial state. The table contains three records:

| ID | Name  | Branch | Marks | Actions       |
|----|-------|--------|-------|---------------|
| 1  | Ravi  | CSE    | 85    | Update Delete |
| 2  | Priya | AIML   | 91    | Update Delete |
| 3  | Kiran | CSE    | 78    | Update Delete |

Output 1: GET request displays all student records.

**Screenshot 2 (Top Right):** After adding a new student. The table now has four records:

| ID | Name  | Branch | Marks | Actions       |
|----|-------|--------|-------|---------------|
| 1  | Ravi  | CSE    | 85    | Update Delete |
| 2  | Priya | AIML   | 91    | Update Delete |
| 3  | Kiran | CSE    | 78    | Update Delete |
| 4  | Anil  | CSE    | 88    | Update Delete |

Output 2: POST request successfully adds a new student record.

**Screenshot 3 (Bottom Left):** After updating a student. The table shows Ravi's marks updated to 90:

| ID | Name       | Branch | Marks | Actions       |
|----|------------|--------|-------|---------------|
| 1  | Ravi Kumar | CSE    | 90    | Update Delete |
| 2  | Priya      | AIML   | 91    | Update Delete |
| 3  | Kiran      | CSE    | 78    | Update Delete |
| 4  | Anil       | CSE    | 88    | Update Delete |

Output 3: PUT request successfully updates Ravi's details.

**Screenshot 4 (Bottom Right):** After deleting a student. The table shows student ID 3 (Kiran) removed:

| ID | Name       | Branch | Marks | Actions       |
|----|------------|--------|-------|---------------|
| 1  | Ravi Kumar | CSE    | 90    | Update Delete |
| 2  | Priya      | AIML   | 91    | Update Delete |
| 4  | Anil       | CSE    | 88    | Update Delete |

Output 4: DELETE request successfully removes student ID 3.

# Algorithm / Procedure

## Step 1

Install Node.js and npm on the system.

## Step 2

Create a new project folder named:

`JSONServerDemo`

## Step 3

Open the terminal in the project folder and initialize the project:

```
npm init -y
```

## Step 4

Install JSON Server: `npm install json-server`

## Step 5

Create a file named: `db.json`

## Step 6

Add sample student data to `db.json`.

## Step 7

Start JSON Server using: `npx json-server --watch db.json --port 3000`

## Step 8

Test the **GET** operation to retrieve all students.

## Step 9

Test the **POST** operation to add a new student.

## Step 10

Test the **PUT** operation to modify an existing student's details.

## Step 11

Test the **DELETE** operation to remove a student.

## Step 12

Verify the changes in `db.json` and record the outputs.

# Program / Code

1. 

Create the following file:

```
{
  "students": [
    {
      "id": "1",
      "name": "Ravi",
      "branch": "CSE",
      "marks": 85
    },
    {
      "id": "2",
      "name": "Priya",
      "branch": "AIML",
      "marks": 91
    },
    {
      "id": "3",
      "name": "Kiran",
      "branch": "CSE",
      "marks": 78
    }
  ]
}
```

## 2. Start JSON Server

```
npx json-server --watch db.json --port 3000
```

```
PS C:\Users\Admin\OneDrive\Desktop\front-end classes> npx json-server --watch db.json --port 3000
--watch/-w can be omitted, JSON Server 1+ watches for file changes by default
JSON Server started on PORT :3000
Press CTRL-C to stop
Watching db.json...
```

(~ ~ ~)

Index:  
<http://localhost:3000/>

Static files:  
Serving ./public directory if it exists

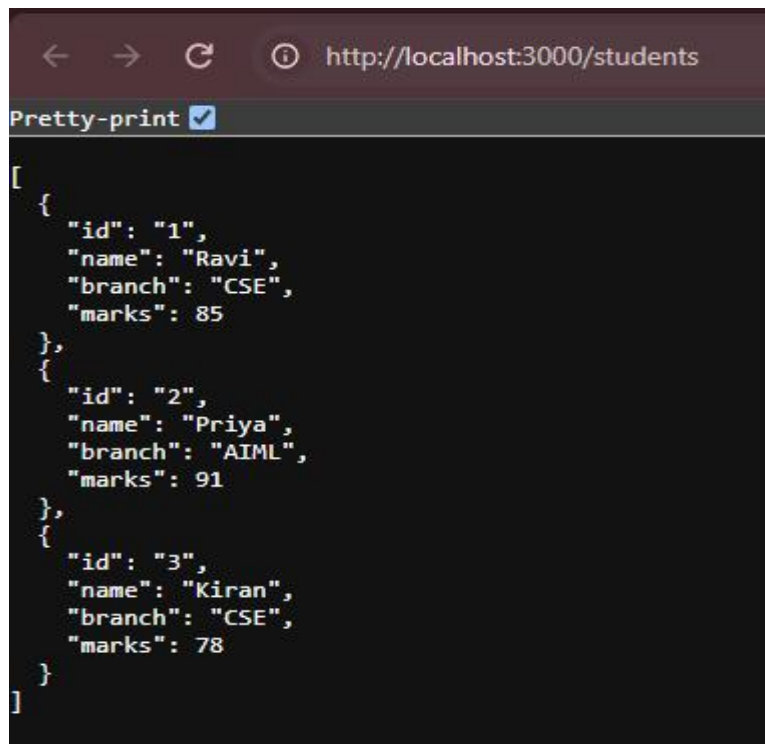
Endpoints:  
<http://localhost:3000/students>

# HTTP Operations

## A. GET Method

Retrieve all students

Method: GET



```
<  >  ↻  ⓘ  http://localhost:3000/students
Pretty-print ☒
[
  {
    "id": "1",
    "name": "Ravi",
    "branch": "CSE",
    "marks": 85
  },
  {
    "id": "2",
    "name": "Priya",
    "branch": "AIML",
    "marks": 91
  },
  {
    "id": "3",
    "name": "Kiran",
    "branch": "CSE",
    "marks": 78
  }
]
```

## B. POST Method

Add a new student

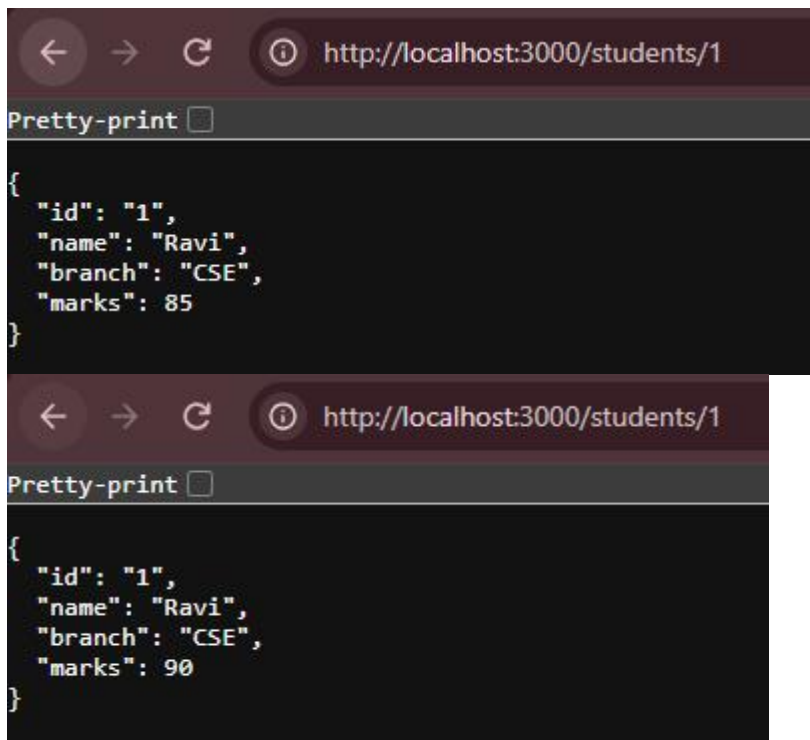
Method: POST

```
{
  "name": "Anil",
  "branch": "CSE",
  "marks": 88
}
```

## C. PUT Method

Update student details

Method: PUT



The details of student with ID 1 will be updated.

## D. DELETE Method

Delete a student

Method: DELETE

<http://localhost:3000/students/3> - The student with ID 3 will be removed.

## Test Cases and Execution Data

Students should record at least the following test cases:

| Test Case | HTTP Method | URL                           | Input / Operation         | Expected Result                    |
|-----------|-------------|-------------------------------|---------------------------|------------------------------------|
| TC01      | GET         | /⑤ ↗ —<br>△ ○ ∅ ↗ ↗ ⑤         | Retrieve all students     | All student records are displayed  |
| TC02      | GET         | /⑤ ↗ ↗ —<br>△ ○ ∅ ↗ ↗ ⑤<br>/1 | Retrieve student ID 1     | Student ID 1 details are displayed |
| TC03      | POST        | /⑤ ↗ ↗ —<br>△ ○ ∅ ↗ ↗ ⑤       | Add Anil, CSE, 88         | New student record is created      |
| TC04      | PUT         | /⑤ ↗ ↗ —<br>△ ○ ∅ ↗ ↗ ⑤<br>/1 | Change Ravi's marks to 90 | Student ID 1 is updated            |
| TC05      | DELETE      | /⑤ ↗ ↗ —<br>△ ○ ∅ ↗ ↗ ⑤<br>/3 | Delete student ID 3       | Student ID 3 is deleted            |

Execution Data

| Operation        | Input                                                                                                                                                                  | Actual Output               | Status      |
|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|-------------|
| GET all students | <br> | 3 student records displayed | <b>Pass</b> |
| GET student      | <br> | Ravi's details displayed    | <b>Pass</b> |
| POST student     | Anil, CSE, 88                                                                                                                                                          | New record added            | <b>Pass</b> |
| PUT student      | ID 1, marks = 90                                                                                                                                                       | Record updated              | <b>Pass</b> |
| DELETE student   | ID 3                                                                                                                                                                   | Record deleted              | <b>Pass</b> |

## Result

Thus, the GET, POST, PUT and DELETE HTTP methods were successfully implemented and tested using JSON Server for managing student records.